

Breaking the Scale Barrier: A Scale-Progressive Water Body Extraction Framework Inspired by Human Cognitive Mechanisms

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Abstract—Water bodies are typical natural features that exhibit significant variations in scale and appearance in high-resolution remote sensing imagery due to the influence of factors such as topography, temperature, sediment content, and aquatic vegetation. Existing deep learning-based water body segmentation networks typically adopt a local sliding-window approach for training and inference. However, this strategy limits the receptive field and hampers the model’s ability to utilize contextual information for identifying large-scale, complex-textured water bodies, often resulting in misclassification. To address these limitations, we propose a scale-progressive prompt water body extraction framework (SPPrompt-Water), inspired by the hierarchical “global-to-local” cognitive mechanisms of human vision. The framework employs progressive prompting to facilitate cross-scale contextual knowledge transfer, thereby enabling high-precision water body extraction. To support this framework, we design an enforced prompt-based parallel simulation training strategy, which leverages cross-scale sample simulation and label decomposition to obtain object-level supervision within the same scene. Through parallel contrastive training, the model learns to perform segmentation with enforced prompt guidance. Extensive experiments conducted on the GID, Complex Water Bodies Segmentation Dataset (CWBSD) and GLH-Water datasets demonstrate that the proposed method effectively balances spatial continuity in large-scale water bodies with boundary precision in small-scale ones. Our method achieves mean Intersection over Union (mIoU) scores of 83.22%, 92.27%, and 86.91% on the respective datasets, outperforming existing approaches by

8.1%, 6.65%, and 6.79%. These results validate the effectiveness of the proposed framework and offer a novel solution for high-precision dynamic monitoring of water bodies using remote sensing imagery. The source code and datasets will be made publicly available at <https://github.com/XinMe/SPPrompt-Water>

Index Terms—Deep learning, high spatial resolution (HSR) imagery, prompt learning, water body extraction.

I. INTRODUCTION

SURFACE water bodies, as a key component of the Earth’s ecosystem, hold significant value in areas such as water resource management, environmental protection, urban planning, and disaster monitoring [1], [2]. Accurate and efficient extraction of water bodies is a critical foundation for informed scientific decision-making [3]. In recent years, remote sensing-based water body extraction techniques driven by deep learning have achieved remarkable progress, significantly improving the precision of water body detection in high-resolution imagery [4]. However, due to factors such as limited training samples, model training and inference strategies, and computational resource constraints [5], current deep learning methods still face challenges in terms of robustness and stability when adapting to the multiscale nature of remote sensing imagery and extracting spatially continuous water bodies in large scenes.

As a typical type of natural land-cover, water bodies exhibit spatial scales and spectral characteristics that are jointly influenced by multiple factors, including terrain variation, water temperature, suspended sediment concentration, and aquatic vegetation distribution [1], [6], [7]. By summarizing existing studies, we identify the following key challenges in deep learning-based water body extraction.

1) **Scale Variability Leads to Contextual Modeling Challenges** [8]. High-resolution remote sensing images often cover extremely large scenes (e.g., on the order of $10\,000 \times 10\,000$ pixels), and a single image may span tens to hundreds of square kilometers. Correspondingly, large water bodies (lakes, coastal water, and oceans) and small ones (ponds, narrow streams) may occupy tens of thousands to only several pixels within an image [6], [9], resulting in substantial differences in shape, size, and geometric morphology (as shown in Fig. 1). However, mainstream deep learning methods typically adopt local sliding-window strategies for both training and inference. Prior studies have shown that such a processing

Received 10 June 2025; revised 24 November 2025 and 15 January 2026; accepted 30 January 2026. Date of publication 6 February 2026; date of current version 20 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U24A20589 and Grant 42301411 and in part by National Key Research and Development Program of China under Grant 2024YFF0617401. (*Corresponding author: Hao Cui.*)

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Digital Object Identifier 10.1109/TGRS.2026.3661185

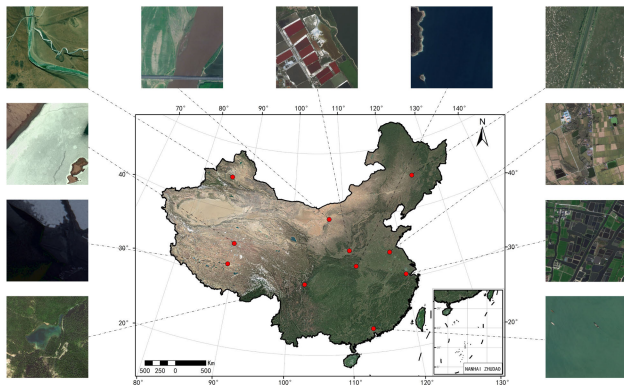


Fig. 1. Comparison of different types of water bodies under the same image size and spatial resolution.

paradigm constrains the model's effective receptive field, making it difficult for the network to capture long-range contextual dependencies within each window [10], [11], particularly in regions with complex textures. Consequently, these limitations can lead to local misclassification errors and cause noticeable mosaic artifacts after stitching the window-wise predictions back into a full-scene result [5], [12].

2) Appearance Variability Leads to Local Feature Extraction Challenges [7], [13]. At a global scale, water bodies typically exhibit distinctive geometric shapes that facilitate their identification. However, at the local scale, their textural characteristics can vary substantially due to factors such as geographic location, seasonal dynamics, sediment concentration, and the distribution of aquatic vegetation. For example, Chen et al. [7] reported that the texture patterns of frozen lakes differ markedly from those of liquid water; similarly, turbid rivers with high sediment loads present significantly different spectral-textural appearances compared with clear water bodies [13].

However, these two challenges often co-occur in practical applications, jointly increasing the difficulty of accurate water-body extraction [14]. Fundamentally, both the contextual gaps caused by scale variations and the local spectral heterogeneity arising from phenotypic differences are closely related to the model's limited ability to leverage contextual information [15]. If cross-scale and cross-region contextual constraints can be incorporated during inference, the model would be able to interpret the overall morphology of water bodies within a broader structural framework, thereby naturally compensating for inconsistencies in local texture and spectral appearance. For example, with access to global spatial context, the model can correctly identify heterogeneous regions within a large water body—such as those affected by ice or algae—as parts of the same class, rather than incorrectly segmenting them into separate components. Therefore, we argue that enhancing contextual modeling capability is an effective approach to simultaneously mitigating both challenges and improving the stability and reliability of water-body extraction in complex scenarios.

We conducted an in-depth analysis of the human cognitive process in remote sensing image interpretation and found that humans typically follow a hierarchical, progressive

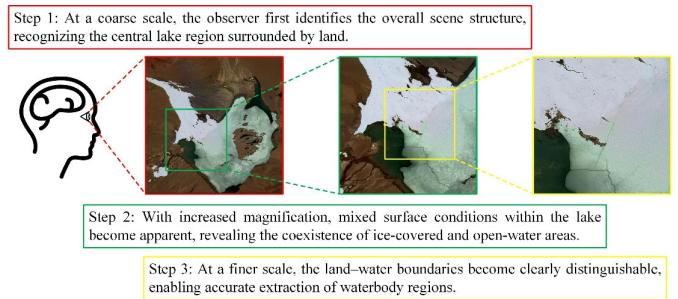


Fig. 2. Cognitive process of human remote sensing image interpretation.

“global-to-local” perception paradigm. This process begins with perceiving the topological contours and spatial distribution of large-scale water bodies through multiscale image pyramids. Subsequently, attention shifts to local regions to analyze spectral and textural details. Finally, coherent water body recognition is achieved by integrating multilevel features. For instance, even when water bodies are in a frozen state and are difficult to identify within small-scale sliding windows, their boundary textures remain distinguishable from land at a global scale, as illustrated in Fig. 2. These global cues can effectively guide subsequent reasoning at finer levels, enabling more precise segmentation at the local scale.

Inspired by this cognitive process, we aim to propose a scale-progressive prompt water body extraction framework (SPPrompt-Water) that simulates the human paradigm of prior-guided global reasoning followed by local refinement. The core innovation of the framework lies in its construction of a hierarchically progressive enforced prompt learning mechanism, which enables effective cross-scale contextual knowledge transfer. Our goal is to overcome the local perception limitations of conventional sliding-window approaches, maintaining spatial continuity for large-scale water bodies while accurately identifying fragmented boundaries and internal heterogeneity of small-scale water bodies. Accordingly, the main objectives of this study are outlined as follows.

- 1) To develop the SPPrompt-Water, which constructs a hierarchically progressive contextual co-optimization mechanism that integrates global semantics with local texture details. It is designed to enhance the model's ability to capture intraclass heterogeneity within water bodies and suppress misclassification in spectrally ambiguous regions.
- 2) To propose an enforced prompt-based supervised learning strategy by decomposing object-level labels into coarse-grained prompt-aware semantic priors and enforcing parallel contrastive training to enhance prompt-guided segmentation.
- 3) To construct a large-scale semantic segmentation dataset for remote sensing water bodies—the complex water bodies segmentation dataset (CWBSD)—which encompasses a diverse range of challenging scenarios, including ice-water mixed lakes, high-sediment rivers, and water surfaces partially obscured by thin clouds.

The rest of the article is organized as follows. Section II introduces the related work in the water extraction field.

Section III details the SPPrompt-Water. Section IV describes the datasets and the experiments. Finally, Section V summarizes the conclusion.

II. RELATED WORK

A. Traditional Remote Sensing Water Body Extraction Methods

Traditional remote sensing methods for water body extraction primarily rely on the strong absorption characteristics of water in the near-infrared (NIR) and shortwave-infrared (SWIR) bands, which result in low reflectance. By exploiting the spectral differences between water and other land covers such as vegetation and soil, these methods achieve classification. Specifically, the features can be divided into two categories: single-band features and multiband spectral relationships. The former directly uses low reflectance values in bands such as NIR (e.g., Landsat B5) to identify water bodies [16], while the latter constructs water indices such as the normalized difference water index (NDWI) and its modified forms (MNDWI) to enhance the contrast between high reflectance in visible bands and strong absorption in NIR/SWIR, thereby improving separability [17], [18], [19].

In terms of implementation, traditional methods mainly include threshold-based approaches, which rely on prior rules, and data-driven classifiers. Threshold methods are computationally simple, but the selection of thresholds depends on human expertise and is prone to errors due to shadows or clouds. Early classifier methods employed statistical models such as maximum likelihood classification, and later evolved to machine learning approaches including support vector machines (SVMs) [20], decision trees [21], and random forests [22].

Traditional methods heavily depend on spectral information and perform well on medium-resolution imagery with relatively complete spectral bands, including SWIR (e.g., Landsat, Sentinel-2). However, high-resolution imagery generally has limited spectral resolution and fewer bands, often lacking SWIR, which leads to spectral confusion between water and dark surfaces such as shadows, thereby limiting the applicability of traditional methods [23], [24].

B. Deep Learning-Based Remote Sensing Water Body Extraction Methods

Driven by advancements in deep learning, the field of semantic segmentation for remote sensing imagery has achieved remarkable progress. In particular, for water body extraction tasks, deep learning methods have demonstrated significant improvements in identifying water bodies under complex scenarios, owing to their powerful capabilities in feature extraction and representation [25].

In recent years, convolutional neural networks (CNNs) have achieved remarkable progress in remote sensing semantic segmentation tasks due to their powerful ability to extract local features, while transformer architectures, known for their superior global modeling capabilities, have gradually become essential tools for interpreting complex scenes.

- 1) CNN-based methods for semantic segmentation perform pixelwise classification of water bodies by leveraging local feature extraction [26], [27]. With the advancement of deep learning, classic architectures such as ResNet [28], U-Net [29], and DeepLab [30] have been widely applied in remote sensing image segmentation, providing a solid technical foundation for water body extraction. For example, DensePPMUNet-a [31] introduces a dense pyramid pooling module to enhance multiscale contextual feature fusion, thereby improving extraction accuracy. Another representative work, the COM framework [32], uses an object-contextual perception network to perform cross-dataset water body extraction, significantly boosting model generalization and applicability across different remote sensing scenes. Additionally, MSResNet [33] incorporates specially designed modules to distinguish water bodies of varying scales and shapes and introduces residual connections to address the difficulty in delineating fragmented water boundaries, thereby enhancing edge recognition performance. However, traditional CNNs are limited by their local receptive fields and struggle to capture the global contextual dependencies that are widely present in remote sensing images. This often leads to susceptibility to spectral noise and intraclass heterogeneity in complex environments [34], [35].
- 2) Transformer-based architectures have revolutionized feature-modeling paradigms [36], [37]. Their self-attention mechanisms enable long-range dependencies between pixels, overcoming the spatial limitations of conventional CNNs [37]. For instance, Ding et al. [38] proposed a hybrid architecture that employs a transformer encoder to capture global contextual features while using a CNN decoder to preserve local details, thereby enhancing the completeness of continuous water body extraction in wide-area imagery. Zhong et al. [39] developed NT-Net, which integrates multilevel transformer modules to suppress background interference and employs boundary enhancement strategies to address the disconnection of small water bodies. Recent research has increasingly focused on heterogeneous feature fusion. For example, Zhang et al. [40] proposed a collaborative Swin transformer-CNN framework that establishes global semantic relationships while preserving local texture details, enabling subpixel-level localization of water boundaries in complex terrains.

In summary, although deep learning-based semantic segmentation methods provide powerful technical support for high-resolution remote sensing water body extraction, their inherent limitations still constrain performance in complex scenarios [6], [38]. First, regarding scale characteristics, mainstream methods generally adopt a “slice-then-process” paradigm [12], [41], which, due to its limited receptive field, struggles to capture long-range contextual dependencies. This often leads to fragmented segmentation of large water bodies and confusion in small water bodies with highly variable morphologies [5]. Second, in terms of feature perception, whether using CNNs or transformer models, the ability to discriminate

local spectral and texture variations remains insufficient when faced with appearance variability caused by ice, snow, or algae [7], [13]. This makes the models susceptible to interference from heterogeneous water regions and background objects, resulting in misclassification. Therefore, exploring a new paradigm that can collaboratively model both global context and local details is crucial to overcoming the current technical bottlenecks [42].

C. Prompt Learning

The fundamental principle of prompt learning lies in encoding task objectives as “prompts” that are input into the model. By leveraging these prompts, the model can activate its prelearned knowledge, enabling it to perform new tasks with no or minimal parameter updates. These prompts can take the form of natural language sentences, trainable vectors, or spatial cues within images. Depending on the content of the prompt, the model adjusts its attention mechanisms and feature processing strategies, guiding it to focus on regions or semantic information relevant to the prompt. This mechanism not only effectively bridges the semantic gap between downstream tasks and pretrained models but also significantly enhances the model’s generalization and transfer capabilities. The theoretical foundation of this paradigm can be traced back to transformative research in the field of natural language processing (NLP). Inspired by large-scale pretrained models such as GPT-3 [43], researchers proposed the use of textual prompts to steer model behavior. This paradigm was subsequently extended to the field of computer vision, where learnable prompt vectors allow models to dynamically reconstruct feature representations based on task context. As a result, models have demonstrated strong performance in visual tasks such as image classification and semantic segmentation [44], [45]. A representative example is the segment anything model (SAM) [46], which establishes a new benchmark in semantic segmentation by incorporating a flexible prompting mechanism and robust zero-shot capabilities.

In the field of intelligent remote sensing interpretation, prompt learning has emerged as a novel paradigm and is gradually being applied to key tasks such as semantic segmentation of geospatial objects and object detection. Current research primarily focuses on the use of visual prompts. For instance, Immanuel and Sinulingga [47], building upon the SegGPT framework [48], proposed an innovative hierarchical prompt learning architecture. By independently optimizing prompt vectors for newly introduced object classes, their approach effectively mitigates the issue of catastrophic forgetting in continual learning scenarios. Similarly, Chen et al. [49] developed RSPrompter, a groundbreaking framework that introduces prompt learning into instance segmentation tasks. Leveraging a self-supervised learning paradigm, RSPrompter adaptively generates visual prompt embeddings enriched with geometric priors. In response to the characteristics of aerial imagery, GeoSAM [50] fine-tunes the SAM by integrating sparse point prompts with dense visual prompts, significantly improving semantic segmentation accuracy for transportation infrastructure. On the other hand, textual prompts are

also beginning to show promise in remote sensing applications. For example, Osco et al. [51] proposed applying the Grounded-SAM framework [52] to remote sensing imagery. By using natural language prompts to guide GroundDINO [53] in object localization, followed by precise segmentation via SAM, the framework enables zero-shot semantic segmentation of remote sensing images based on textual descriptions.

In summary, prompt learning has demonstrated superior feature extraction capabilities in computer vision tasks. In recent years, this approach has been gradually introduced into the field of intelligent remote sensing interpretation, with several studies proposing adaptive prompt learning frameworks for tasks such as land-cover classification and instance segmentation. Building upon the strengths of prompt learning, we conduct research on remote sensing-based waterbody interpretation, aiming to enhance the accuracy of waterbody extraction and improve boundary delineation performance.

III. METHODOLOGY

We simulate the hierarchical and progressive cognitive mechanism of human vision—from global perception to local refinement—and propose a scale-progressive, prompt-guided framework for waterbody extraction. As illustrated in Fig. 3, this framework consists of two main components: network architecture design and a training–inference strategy. Architecturally, we develop a novel network named SPPrompt-Water, which incorporates a prompt guidance module to transfer spatial-semantic information across different hierarchical levels. This facilitates layerwise refinement of segmentation results and enables accurate delineation of complex waterbody structures. In terms of strategy, we introduce an enforced prompt-based parallel simulation training approach that enhances the model’s prompt guidance capability by integrating label decomposition and perturbation simulation mechanisms. During inference, we adopt a scale-progressive reasoning mechanism, which follows a closed-loop optimization process of “downsampling–prompting–upsampling,” achieving fine-grained segmentation of waterbody regions in a progressive manner from global context to local detail.

A. SPPrompt-Water Net

The SPPrompt-Water Net is responsible for encoding both the image and the prompts at each layer, facilitating guided fusion. By integrating semantic guidance and detail enhancement, the network improves segmentation performance in complex scenes. The overall architecture comprises a prompt guidance module and a detail refinement module, as illustrated in Fig. 4. The specific design is as follows.

1) *Prompt Guidance Module*: The primary function of the prompt guidance module is to fuse the low-resolution segmentation results from the previous layer with the higher-resolution image features at the current layer. The core idea is to treat the low-resolution prompt map from the previous layer as an analogy to the global perception process in human vision, which provides a coarse estimation of the target’s

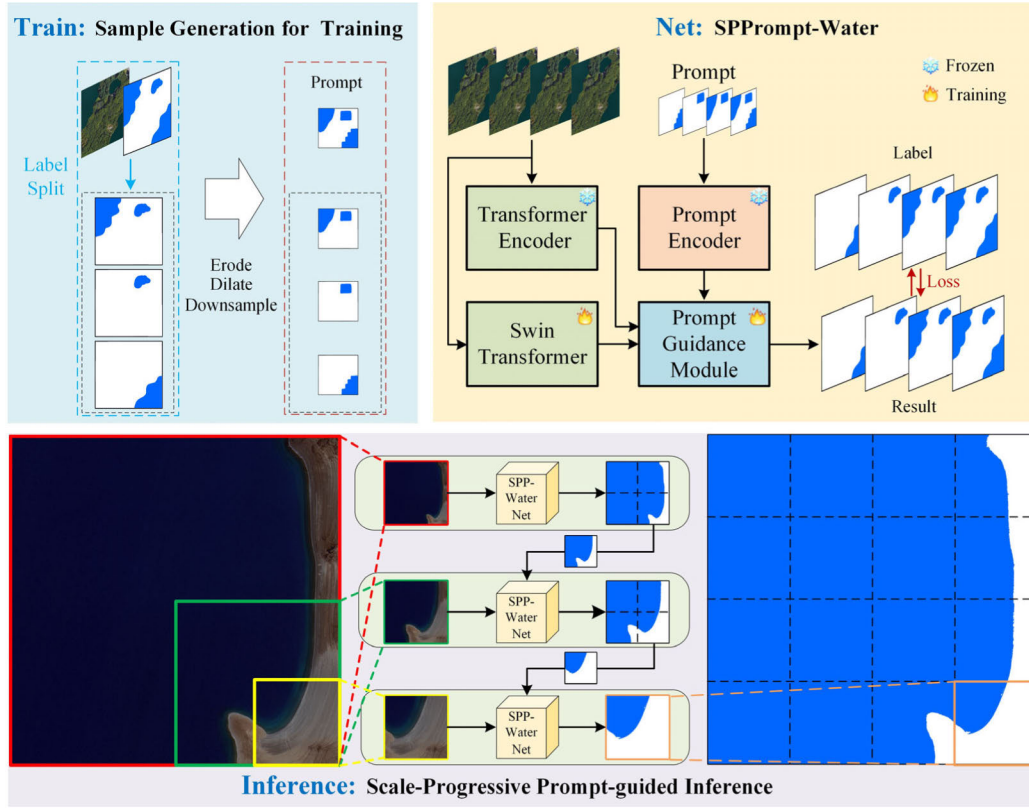


Fig. 3. Overall architecture of the proposed framework.

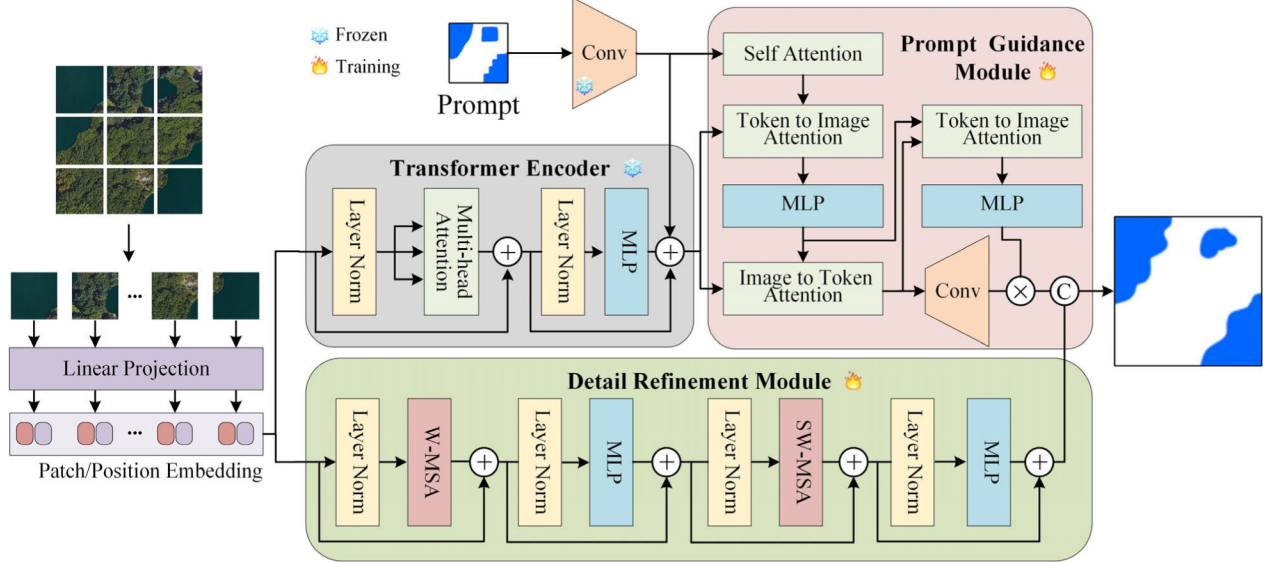


Fig. 4. SPPrompt-Water Net.

spatial extent. In contrast, the high-resolution image at the current layer offers rich detail information to further refine boundary representations based on the coarse outline. This process is conceptually similar to a pyramid-based multiscale processing strategy, where global structures are first captured at lower resolutions, and subsequently, higher-resolution layers are guided to extract local fine-grained features, ultimately leading to more accurate segmentation results.

The detailed workflow is as follows. Given a prompt image $I_p \in \mathbb{R}^{H/2 \times W/2}$, we first obtain the prompt embedding features E_p through convolution. A learnable output vector q is then introduced into the prompt embedding to construct the enhanced prompt representation $Q_p = [E_p; q]$. Based on this, prompt guidance module first performs self-attention within the prompt to capture the global structural relationships among the prompt tokens. Next, a bidirectional cross-modal

attention mechanism is established: the token-to-image attention transfers positional and structural prior information from the prompt tokens to the image features, guiding the backbone network to focus on potential target regions; conversely, the image-to-token attention uses image semantics to adaptively modulate the prompt features, allowing the prompt information to adjust to the current scene distribution. After these two information flows pass through three layers of MLP for further abstraction, a context-aware fused representation is obtained

$$P_{\text{fuse}} = \text{MLP}^3(\text{AttnFuse}(Q_P, F_{\text{trans}})) \quad (1)$$

where $\text{AttnFuse}(\cdot)$ represents the integrated computation of the aforementioned self-attention and bidirectional cross-modal attention. Finally, after the image feature K_{img} is upsampled via transposed convolution, it is multiplied elementwise with P_{fuse} to obtain the water foreground probability map as follows:

$$F_{\text{prompt}} = \text{Conv}(K_{\text{img}}) \odot P_{\text{fuse}}. \quad (2)$$

This module leverages the bidirectional guidance between prompt priors and image features to effectively enhance the model's discriminative capability in complex backgrounds.

We adopt the pretrained SAM [46] model as the encoder of the transformer encoder to fully leverage its powerful segmentation capabilities. To preserve its original feature extraction performance, we freeze the parameters of the transformer encoder during training, keeping its weights unchanged without fine-tuning.

2) *Detail Refinement and Feature Fusion:* Although the transformer encoder effectively captures global semantic information, it primarily serves the prompt guidance module and is thus susceptible to the influence of large-scale responses in the prompted regions. This can lead to the underrepresentation of fine-grained, small-scale water bodies. To address this limitation, we introduce the Swin transformer [54] as a parallel branch to enhance the extraction of local detail features.

The Swin transformer employs window-based multihead self-attention (W-MSA) to model local features. Unlike ViT [37], which performs self-attention globally, W-MSA restricts attention computation to nonoverlapping local windows, thereby reducing computational complexity. However, this limitation also constrains information exchange across different windows. To address this issue, the Swin Transformer introduces the SW-MSA mechanism in adjacent layers. By shifting window positions between layers, SW-MSA facilitates cross-window feature fusion, thereby enhancing the model's capability in fine-grained representation learning.

At the final stage of the network, in order to further integrate local detail features with prompt-guided information, we concatenate the feature representation F_{swin} , extracted by the Swin transformer, with the prompt-guided features F_{prompt} , output by the prompt guidance module, along the channel dimension to obtain the final fused representation Y , as formulated in the following equation:

$$Y = \text{Concat}(F_{\text{swin}}, F_{\text{prompt}}). \quad (3)$$

To address the significant class imbalance in this task (i.e., background pixels vastly outnumber target pixels), we employ a hybrid loss function combining dice loss [55] and focal loss

[56]. This choice is motivated by the complementary strengths of the two losses in handling class imbalance. Dice loss tackles macroscopic class imbalance by directly optimizing the overlap of the predicted segmentation regions and is defined as follows:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum y_{\text{true}} y_{\text{pred}} + \epsilon}{\sum y_{\text{true}} + \sum y_{\text{pred}} + \epsilon} \quad (4)$$

where y_{true} and y_{pred} denote the ground-truth labels and predicted probabilities, respectively, and ϵ is a small smoothing term to prevent division by zero. By measuring the overlap of the entire segmentation region, this loss is insensitive to the number of foreground pixels, thereby effectively improving the segmentation performance of small targets.

However, dice loss may lack sufficient focus when dealing with a large number of simple background pixels. Therefore, we introduce focal loss to concentrate on hard-to-classify pixels at a more fine-grained level. It dynamically down-weights easily classified samples through a modulating factor and is defined as follows:

$$\mathcal{L}_{\text{Focal}} = -\alpha (1 - y_{\text{pred}})^\gamma \log(y_{\text{pred}}) \quad (5)$$

where α is the balancing factor and γ is the focusing parameter. In this work, we set $\gamma = 2$ and $\alpha = 1$. This design ensures that the model continuously pays attention to more challenging samples during training, thereby further improving segmentation accuracy in complex backgrounds. By combining the strengths of both losses, we define the final loss function as their weighted sum

$$\mathcal{L}_{\text{Total}} = \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{Focal}} \mathcal{L}_{\text{Focal}}. \quad (6)$$

Dice loss ensures the overall geometric integrity of the segmented targets, while focal loss refines pixel-level classification boundaries. We expect that the synergistic effect of the two losses will jointly enhance the model's robustness and overall performance in scenarios with class imbalance.

B. Prompt-Guided Training and Inference Method

To address the limitations of conventional image-label supervision paradigms, in which models passively fit the data distribution and lack dynamic semantic guidance, we propose a prompt-guided training and inference method. During training, we construct an enforced prompt-based parallel simulation framework that explicitly enhances the association between prompt vectors and target regions by decomposing label regions and prompt samples for parallel training. In the inference phase, we introduce a scale-progressive prompting mechanism that follows a closed-loop optimization process consisting of downsampling, prompting, and upsampling. This mechanism integrates multiscale feature information, allowing coarse-grained prompts to progressively guide fine-grained segmentation. It enables hierarchical and refined delineation of water bodies, from global coherence to local detail, effectively alleviating the spatial fragmentation problem caused by patch-based processing in traditional approaches.

1) Enforced Prompt-Based Parallel Simulation Training:

The conventional image-label supervision paradigm relies on supervised training for the model to learn the mapping from images to annotations, which essentially results in passive fitting of the data distribution. In this study, we propose the enforced prompt-based parallel simulation-training framework. Its core idea is to provide only partial supervisory information for the sample images and, through prompt learning and contrastive training, enable the model to acquire the ability to perform segmentation guided by prompts, achieving a “segment where pointed” effect. The overall framework is summarized in Algorithm 1.

As shown in Fig. 5, during the training stage, we automatically construct prompt samples from the annotated data through a three-step process.

- 1) *Label Topological Decomposition*: We first perform connected-component analysis on water body labels, decomposing complex whole-water polygons into multiple spatially independent subregions to obtain object-level supervision.
- 2) *Morphological Perturbation (Erosion/Dilation)*: To simulate potential uncertainties in real prompts, each subregion is randomly eroded or dilated (with the structural element size randomly chosen between 3 and 10), generating noisy prompt maps that enable the model to handle imprecise prompts.
- 3) *Downsampling*: To create coarse-grained prompts compatible with the hierarchical network, the perturbed prompt maps are downsampled to obtain low-resolution prompts, which serve as guidance for lower-level network inputs. Through these three steps, we generate a set of enhanced prompt samples corresponding to the original images, along with their region-level supervisory labels.

Let the original label image be $L \in \{0, 1\}^{H \times W}$. By applying the topological decomposition function $T(\cdot)$, an initial set of independent subregions $T(L)$ can be obtained. Based on this, the RandomMerge function randomly samples from these subregions and merges the sampled subregions using a pixelwise union, thereby generating a set of prompts R with varying coverage

$$R = \text{RandomMerge}(T(L)) = \{R_1, R_2, \dots, R_N\}. \quad (7)$$

Subsequently, as shown in (8), each image R_j in the prompt set R undergoes erosion $E()$, dilation $D()$, and downsampling $\text{Downsample}()$ operations

$$P_j = \text{DownSample}(D(E(R_j))) \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2}} \quad (8)$$

where the kernel size for erosion and dilation is randomly selected between 3 and 10, the number of iterations set to 1.

During the training phase, images and prompt patches are jointly input into the SPPrompt-Water Net to achieve guided feature interaction and water body region extraction. The prompt patches serve as spatial prior knowledge to construct spatially constrained supervision, enhancing the model's perception of designated regions. Specifically, for the same remote sensing image, multiple different prompt maps are constructed (e.g., P_2 , P_3 , and P_4 in Fig. 5), each covering different water

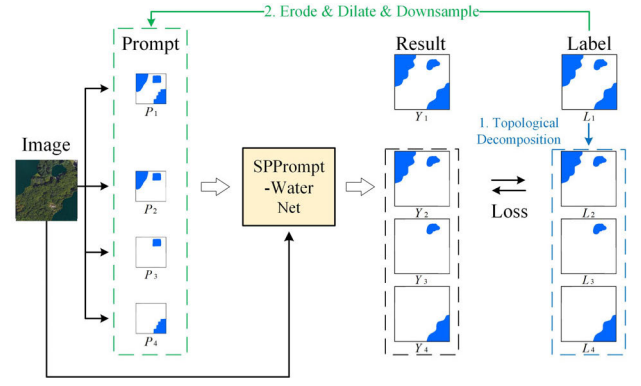


Fig. 5. Illustration of enforced prompt-based parallel simulation training.

Algorithm 1 Enforced Prompt-Based Parallel Simulation Training

Input: Training image set I , label set L , number of iterations T
Output: Trained model parameters Θ

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1: for  $t = 1$  to  $T$  do
2:   Sample a mini-batch ( $I_{\text{batch}}, L_{\text{batch}}$ ) from the dataset
3:   for each image-label pair ( $I, L$ ) in ( $I_{\text{batch}}, L_{\text{batch}}$ ) do
4:     # 1. Label decomposition and prompt generation
5:      $R = \text{RandomMerge}(\text{TopologicalDecomposition}(L))$  # Eq.(7)
6:      $\text{Prompts}_{\text{set}} = \emptyset$ 
7:      $\text{Label}_{\text{set\_set}} = \emptyset$ 
8:     for each decomposed region  $R_j$  in  $R$  do
9:        $P_j = \text{DownSample}(\text{Dilation}(\text{Erosion}(R_j)))$  # Eq.(8), prompt map
10:       $L_j = R_j$  # Guidedlabel
11:      Add ( $P_j, L_j$ ) into  $\text{Prompts}_{\text{set}}$  and  $\text{Label}_{\text{set}}$ 
12:   end for
13:   # 2. Parallel simulation training
14:   for each prompt-label pair ( $P, L_{\text{guide}}$ ) in ( $\text{Prompts}_{\text{set}}, \text{Label}_{\text{set}}$ ) do
15:      $Y_{\text{pred}} = \text{SPPrompt-Water-Net}(I, P)$  # Forward pass
16:      $\text{Loss} = \text{ComputeLoss}(Y_{\text{pred}}, L_{\text{guide}})$  # Simplifiedloss
17:     Backpropagate and accumulate gradients
18:   end for
19:   Update model parameters  $\Theta$  using accumulated gradients
20: end for
21: end for
22: return  $\Theta$ 

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body regions within the image, thereby forming corresponding region-guided labels (e.g., L_2 , L_3 , and L_4 in Fig. 5). Based on the correspondence between each prompt and its label, the network performs local supervised learning under different prompt conditions, progressively guiding the establishment of an explicit association from prompts to target responses. This mechanism enforces the model to learn texture features within the prompt range for region-focused constraints and to apply feature activation penalties to nonprompt regions for background suppression.

2) *Scale-Progressive Prompt-Guided Inference*: In this section, we design a scale-progressive prompt inference mechanism that achieves gradual refinement from global structures to local boundaries. This method constructs a multiscale feature pyramid based on cascaded downsampling, using coarse-grained prompts as high-level priors to progressively guide the optimization of low-level local features. Through a closed-loop process of downsampling, prompting, and upsam-

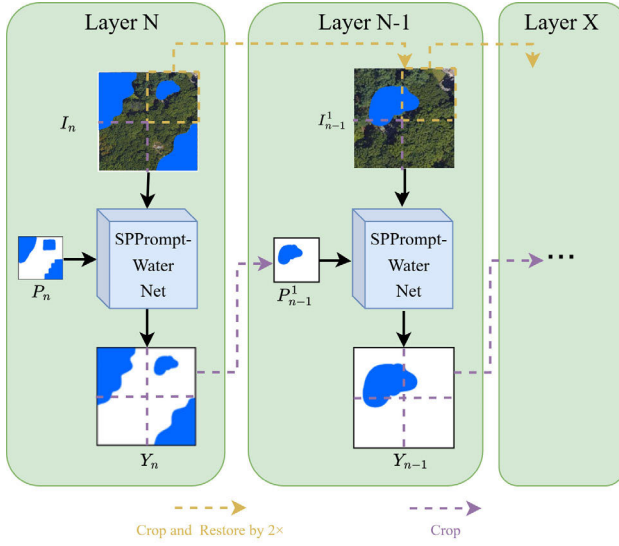


Fig. 6. Illustration of the scale-progressive prompt inference mechanism.

pling, multiscale semantic collaborative inference is realized, effectively enhancing the continuity expression and boundary delineation of complex water body structures.

Given the full-scene remote sensing image I_0 , a multiresolution feature pyramid is constructed by cascaded downsampling at a rate of 2^n per level. As illustrated in Fig. 6, the highest downsampling level is defined as Layer N , with its resolution corresponding to $I_n = I_0 \times 1/2^n$. At this level, the image $I_n \in \mathbb{R}^{H \times W}$ is obtained, along with the corresponding coarse-grained prompt $P_n \in \mathbb{R}^{H/2 \times W/2}$. At Layer N , the initial segmentation result $Y_n \in \mathbb{R}^{H \times W}$ is first generated through the SPPrompt-Water Net

$$Y_n = \text{SPPrompt}(I_n, P_n). \quad (9)$$

The initial result Y_n is divided along the spatial dimension into four subregions $\{P_{n-1}^i\}_{i=1}^4 \in \mathbb{R}^{H/2 \times W/2}$, which serve as prompt embeddings $P_{n-1}^i \in \mathbb{R}^{H/2 \times W/2}$ for layer $N-1$. Simultaneously, the image $I_n \in \mathbb{R}^{H \times W}$ is partitioned in the same spatial manner and then upsampled by a factor of two to obtain the aligned layer $N-1$ images $I_{n-1}^i \in \mathbb{R}^{H \times W}$. Subsequently, more refined segmentation results $Y_{n-1}^i \in \mathbb{R}^{H \times W}$ are generated through SPPrompt-Water Net. This top-down process is repeated at each layer with the same operations until the original resolution is restored at layer $N=0$. As shown in (10), at layer 0, the images are restored to the original resolution, and all subregions $\{Y_0^i\}_1^{4^N}$ are concatenated according to their spatial positions to reconstruct the complete high-resolution segmentation result Y_0

$$Y_0 = \text{mosaic} \left(\bigcup_1^{4^N} \text{PPrompt}(I_0^i, P_0^i) \right). \quad (10)$$

The above process is summarized in Algorithm 2.

Taking the GID dataset as an example, its original image resolution is 7300×6908 pixels. To meet the scale consistency requirements of the hierarchical segmentation network, the image is first padded with zeros to expand it to 8192×8192 pixels. Based on the ratio between the image size and the minimum sub-image scale, the number of hierarchy levels was set to 3. Finally, at level 0, the complete

segmentation result was reconstructed by spatially stitching together 4^3 subimages, achieving a high-precision prediction output consistent with the original image size.

To accommodate scenarios where explicit prompts are unavailable during inference, we introduce a “default prompt token” at the highest scale layer (Layer N), which is implemented as a zero-valued tensor and is shared across all images and scenes. When no prompt image is provided, this token serves as a placeholder input to the prompt module, enabling the network to operate normally under a “no-prompt” condition. Although it does not carry any prior information about the scene, it preserves the structural consistency required by the prompt mechanism. Subsequently, the model uses the prediction from Layer N as the prompt for Layer $N-1$, and this process is repeated across subsequent layers. Through this top-down, progressive prompt mechanism, the model can generate increasingly refined pseudo-prompts at each layer, ultimately producing reliable water body segmentation results.

Algorithm 2 Scale-Progressive Prompt-Guided Inference

Input: Full-scene image I_0 , model SPPrompt-Water-Net, pyramid levels N

Output: Final segmentation Y_0

```

1: #Build image pyramid
2: for  $n = N$  down to 0 do
3:    $I_n = \text{Downsample}(I_0, \text{factor} = 2^n)$ 
4: end for
5: # Top-level coarse segmentation
6:  $P_N = \text{InitializePrompt}(\text{Size}(I_N))$ 
7:  $Y_N = \text{SPPrompt-Water-Net}(I_N, P_N)$ 
8: #Progressive top-down refinement
9: for  $n = N-1$  down to 0 do
10:   $P_N = \text{Split}(Y_{n+1})$  # 4 sub-prompts
11:   $I_{n\_sub} = \text{Split}(I_n)$  # 4 sub-images
12:
13:  for  $i = 1$  to 4 do
14:     $Y_{ni} = \text{SPPrompt-Water-Net}(I_{n\_subi}, P_{ni})$ 
15:  end for
16:
17:   $Y_n = \text{Mosaic}(\{Y_{ni}\})$  # merge sub-results
18: end for
19: return  $Y_0$ 

```

At each scale level, the low-resolution prediction from the previous layer is directly used as the prompt input for the next layer, forming a “coarse-to-fine” hierarchical prompt chain. The coarse-scale prompts provide the overall structure and large-scale water distribution, while the fine-scale modules iteratively refine local boundaries under the guidance of the prompts, achieving progressive refinement from global to local. This closed-loop, progressive inference mechanism significantly enhances both the spatial continuity of large-scale water bodies and the detailed delineation of small-scale boundaries.

IV. EXPERIMENTS

A. Datasets

To validate the generalization ability of the SPPrompt-Water network, we conducted experiments on two datasets: the GID dataset and the CWBSD dataset.

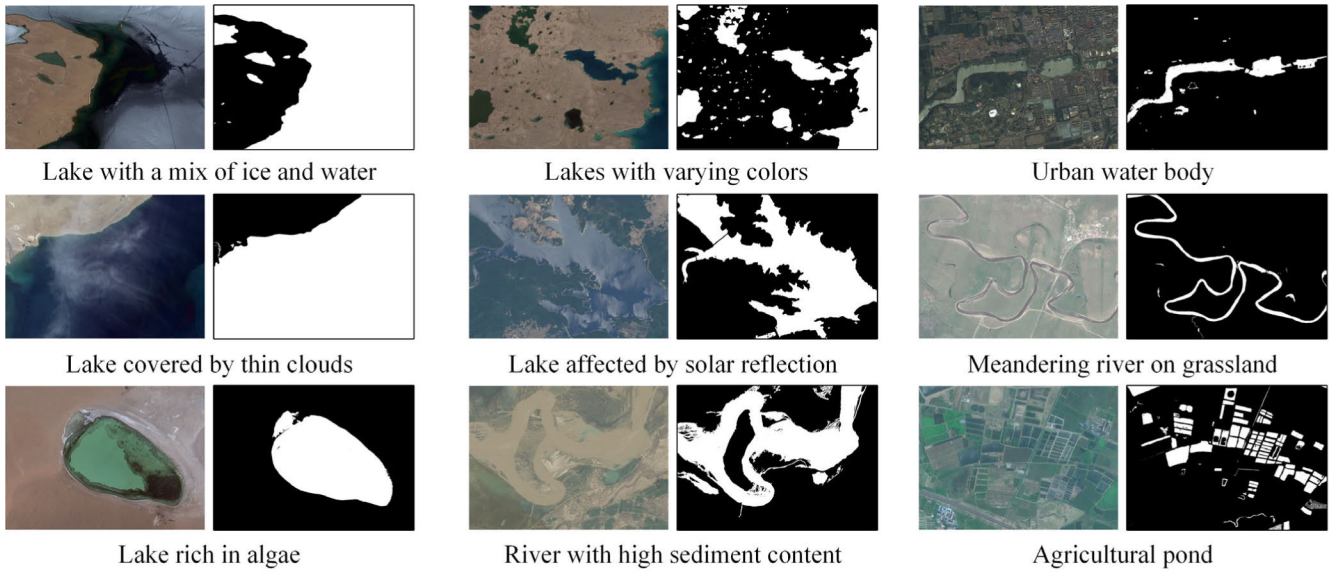


Fig. 7. Illustration of different types of water bodies in the CWBSD dataset.

1) *GID*: The GID dataset is constructed using imagery from the Gaofen-2 (GF-2) satellite and contains a total of 150 scenes with an original resolution of 7300×6908 pixels. It is primarily designed for land cover classification tasks. The original dataset includes annotations for five land cover classes: built-up, farmland, forest, meadow, and water. To meet the requirements of our method, we retained only the water class and divided the images into smaller patches of 1024×1024 pixels to ensure that the model could more effectively learn water-related features. The test set defined by the original authors of the GID dataset was used for evaluation, while the remaining images were allocated for training and validation, ensuring the comparability and validity of experimental results.

2) *CWBSD*: To further validate the effectiveness of SPPrompt-Water Net, we constructed a dataset encompassing a wide range of complex water body scenarios. This dataset includes various challenging water types, such as frozen lakes, ice-water mixed lakes, algae-rich waters, lakes partially covered by thin clouds, rivers with high sediment concentrations, and water surfaces affected by solar reflections, as illustrated in Fig. 7. Additionally, the dataset contains small-scale artificial water bodies such as man-made channels and ponds, ensuring a diverse representation of water types. The dataset comprises a total of 360 remote sensing images with variable sizes, ranging from 1665×1121 – $18\,722 \times 11\,612$ pixels, and spatial resolutions between 0.8 and 2 m. Each image contains three spectral bands (RGB) and is sourced from the Gaofen-1 (GF-1) and GF-2 satellites. During training, we uniformly cropped the images into 1024×1024 patches. To ensure class balance, we removed patches that did not contain any water bodies. The dataset was then split into 60% for training, 20% for validation, and 20% for testing. The test set is made publicly available to facilitate reproducibility and further research.

3) *GLH-Water*: The GLH-water dataset is a large-scale, ultrahigh-resolution satellite imagery dataset designed for

global surface water detection. It contains a total of 250 satellite images, each with a resolution of $12\,800 \times 12\,800$ pixels and a spatial resolution of 0.3 m. The images are sourced from the Google Earth platform, covering multiple regions worldwide (excluding Antarctica) and spanning the period from 2011 to 2022 [57].

B. Evaluation Indexes

To quantitatively evaluate the performance of different methods, we compared the predicted masks with the ground truth and adopted three commonly used metrics: overall accuracy (OA), mean intersection over union (mIoU), and *F1*-score. The formulas for these metrics are defined as follows:

$$\begin{aligned} OA &= \frac{TP + TN}{TP + TN + FP + FN} \\ mIoU &= \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i + FN_i} \\ F1 &= \frac{2TP}{2TP + FP + FN} \end{aligned} \quad (11)$$

where TP, TN, FP, and FN represent the number of true water pixels, true background pixels, false water pixels, and false background pixels, respectively. N denotes the total number of classes, and i indicates the i th class. In this study, N is set to 2.

C. Comparative Experiments

To evaluate the effectiveness of the proposed SPPrompt-water method in waterbody extraction tasks, we conducted comparisons with seven representative deep learning-based waterbody extraction methods: U-Net [29], Swin transformer [54], QTUNet [10], MECNet [11], MSLANet [25], MSResNet [33], and SkySense++ [58]. These methods were tested on the GID, CWBSD, and GLH-Water datasets.

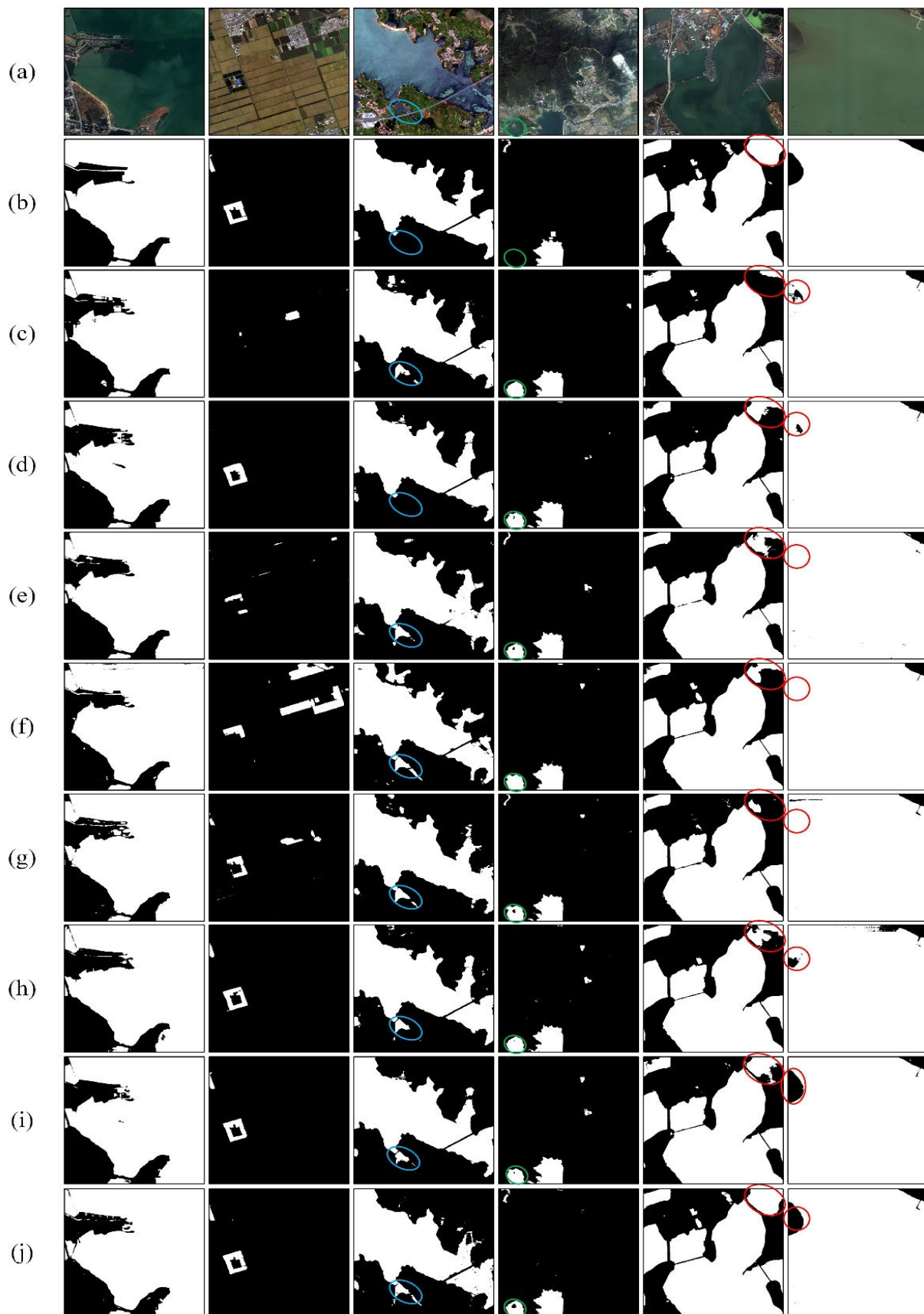


Fig. 8. Visual results of different methods on GID. (a) Image. (b) Label. (c) U-Net. (d) Swin Transformer. (e) QTNUnet. (f) MECNet. (g) MSLANet. (h) MSResNET. (i) SkySense++. (j) SPPrompt-Water (ours).

During inference, we introduce the proposed scale-progressive prompt-guided inference mechanism. In contrast, baseline methods, which do not incorporate a prompt-guided structure, still rely on a sliding-window inference strategy, where the full image is divided into patches for prediction and then reconstructed into a complete result. To ensure fairness, all quantitative metrics for the compared methods are calculated and evaluated based on the reconstructed full-scene predictions.

1) *GID*: As shown in Table II, on the GID dataset, the proposed SPPrompt-Water method achieved state-of-the-art segmentation performance, outperforming all comparative methods across multiple evaluation metrics. Specifically, it reached an OA of 99.09%, with an mIoU of 83.22%, and an *F1*-score of 85.92%. Although most methods reported high OA values (all exceeding 96%), this is largely due to the dominance of background and target categories in the test set, which limits the sensitivity of OA in reflecting true model discrimination—especially in the presence of minimal boundary complexity. In contrast, mIoU and *F1*-score provide more reliable evaluations of a model's ability to accurately identify waterbodies. For example, in terms of mIoU, SPPrompt-Water outperforms the second-ranked SkySense++ by 8.1% points (83.22% versus 75.12%), and in *F1*-score, it exceeds Swin transformer by 7.07% points (85.92% versus 78.85%), demonstrating its superior accuracy in target localization and boundary delineation. These results indicate that SPPrompt-Water not only distinguishes water bodies from the background accurately but also possesses stronger representational and discriminative capabilities when handling complex boundaries and fine structures.

As shown in the comparison results in Fig. 8, SPPrompt-Water demonstrates superior performance in preserving waterbody boundary integrity and capturing fine details, effectively reducing shape distortion and misclassification. In complex scenarios where waterbodies are interwoven with farmland, models such as QTNet, MECNet, and MSLANet exhibit notable confusion between farmland and pond categories. In contrast, SPPrompt-Water achieves higher accuracy in both regional segmentation and class discrimination. In the blue-highlighted area of Fig. 8, SPPrompt-Water successfully extracts water branches that are not labeled in the ground truth, whereas Swin transformer fails to detect this region. Although U-Net, QTNet, and MSResNet do identify parts of the waterbody, their segmentation results suffer from blurred boundaries and noticeable shape distortions. Similarly, in the green-highlighted area, a small pond in the lower-left corner is not annotated in the ground truth, yet all methods detect it. Among them, SPPrompt-Water achieves more precise edge extraction, especially in delineating nonwater regions within the pond, effectively distinguishing waterbodies from the background and thus improving segmentation accuracy and spatial consistency. These results confirm that SPPrompt-Water possesses enhanced boundary awareness and achieves higher segmentation accuracy in complex waterbody scenarios, further validating its effectiveness in waterbody extraction tasks.

TABLE I
QUANTITATIVE COMPARISON RESULTS OF GID

Methods	OA	mIoU	F1score
U-Net	96.75%	71.43%	75.93%
SwinTransformer	97.77%	75.00%	78.85%
QTNet	96.01%	68.59%	73.08%
MECNet	97.03%	72.10%	76.29%
MSLANet	96.85%	71.81%	73.08%
MSResNET	98.04%	74.70%	78.58%
SkySense++	97.91%	75.12%	78.64%
SPPrompt-Water (ours)	99.09%	83.22%	85.92%

TABLE II
QUANTITATIVE COMPARISON RESULTS OF CWBSD

Methods	OA	mIoU	F1score
U-Net	92.64%	80.01%	84.37%
SwinTransformer	96.10%	85.29%	88.47%
QTNet	89.76%	75.17%	79.47%
MECNet	91.09%	78.58%	82.40%
MSLANet	92.87%	78.58%	82.96%
MSResNet	97.09%	85.62%	88.99%
SkySense++	96.21%	85.43%	88.51%
SPPrompt-Water (ours)	99.51%	92.27%	93.97%

2) *CWBSD*: As shown in Table II, the proposed SPPrompt-Water method achieves the best performance on the CWBSD dataset, with an OA of 99.51%, mIoU of 92.27%, and *F1*-score of 93.97%, significantly outperforming all other comparison methods. Compared with the second-best performing model, MSResNet, SPPrompt-Water improves OA by 2.42%, mIoU by 6.65%, and *F1*-score by 4.98%. These improvements indicate that the model possesses a stronger discriminative capability along class boundaries, effectively reducing both false positives and false negatives in waterbody edge regions. Furthermore, it enhances recognition accuracy in small-scale or morphologically irregular waterbodies, thereby improving the model's overall spatial consistency and semantic completeness. These results demonstrate that by introducing a prompt guidance module, SPPrompt-Water effectively integrates multiscale spatial contextual information, significantly enhancing the model's ability to identify waterbodies of various scales and irregular shapes.

Through comparative experiments on representative scenes shown in Fig. 9, we conducted a visual evaluation of the semantic segmentation performance of each model. When the water surface is densely covered with algae accompanied by significant color variation, the baseline models tend to misclassify these areas due to an over-response to the algae texture, mistakenly identifying clusters of algae as water boundaries. In contrast, SPPrompt-Water leverages a multiscale prompting mechanism to effectively distinguish between algal biomass and true water regions, demonstrating superior robustness to such interference. In scenes where there is still a certain degree of algal interference but a distinct boundary between land and lake exists, the overall segmentation performance across models is more satisfactory and the differences among models are less noticeable. This may be because models can more

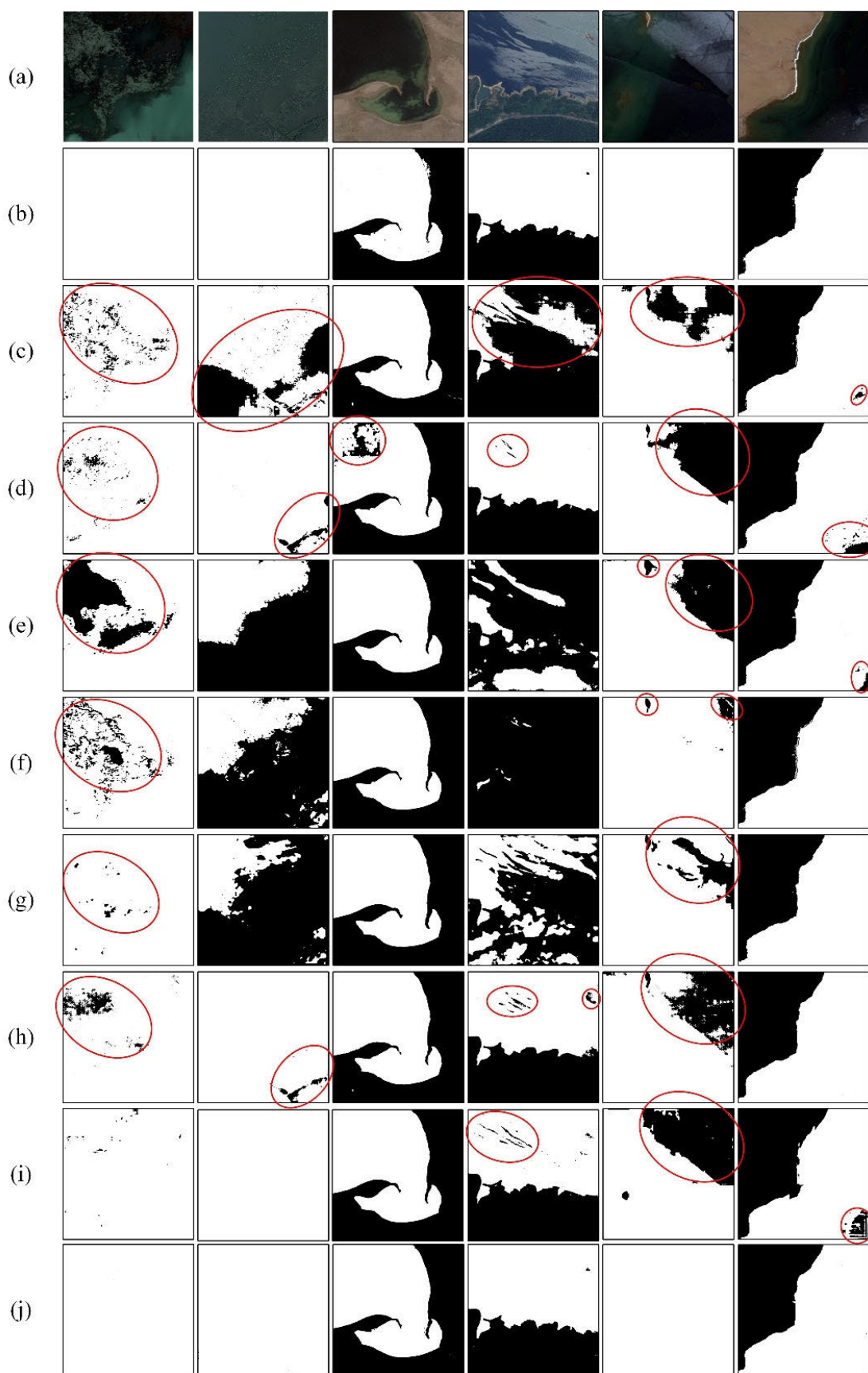


Fig. 9. Visual results of different methods on CWBSD. (a) Image. (b) Label. (c) U-Net. (d) Swin Transformer. (e) QTNUnet. (f) MECNet. (g) MSLANet. (h) MSResNET. (i) SkySense++. (j) SPPrompt-Water (ours).

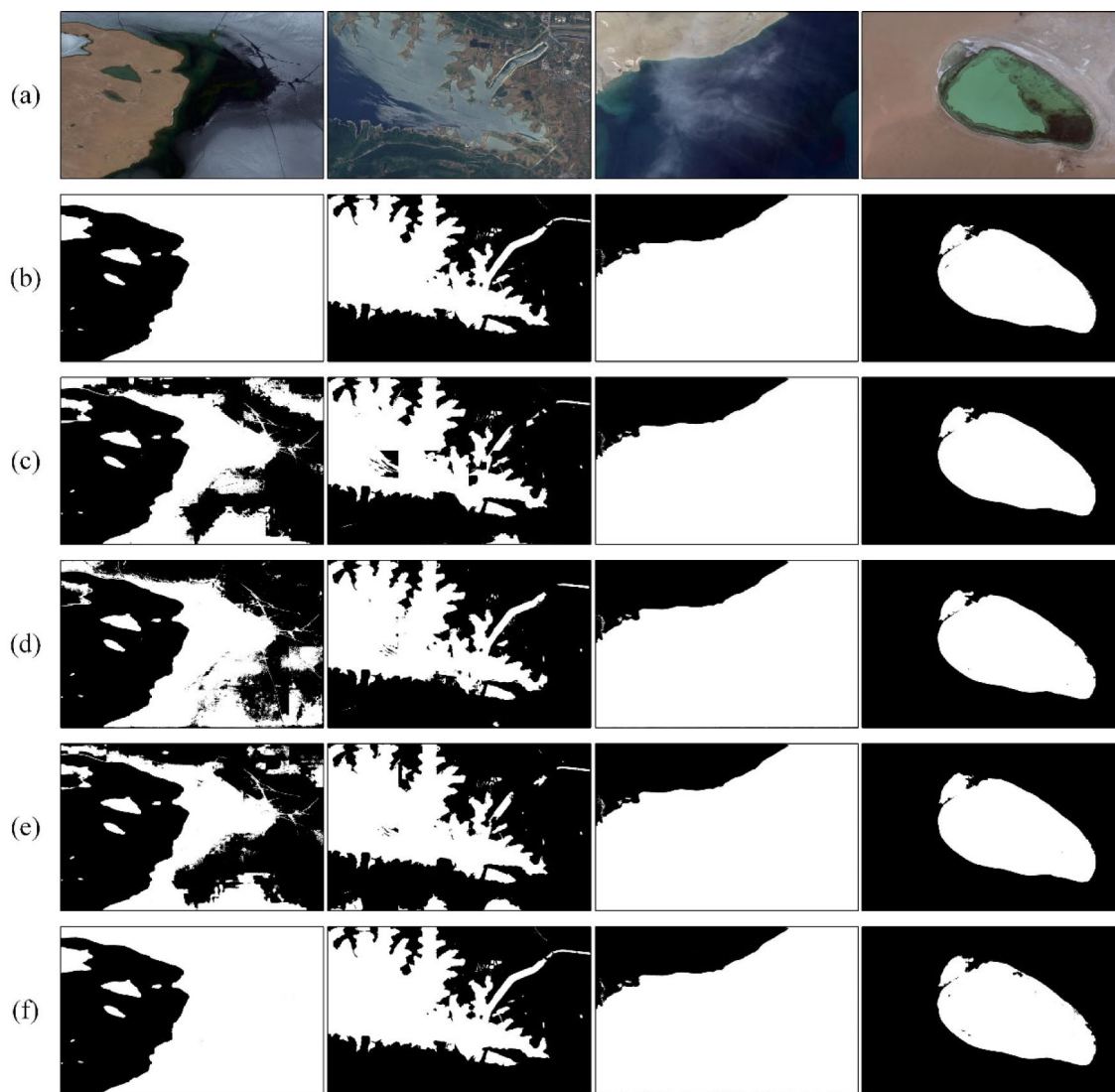


Fig. 10. Visual results of different methods on full-scene images from the CWBSD. (a) Image. (b) Label. (c) Swin transformer. (d) MSResNET. (e) SkySense++. (f) SPPrompt-Water (ours).

easily learn discriminative features from objects with strong color contrast, resulting in accurate semantic segmentation. In comparison, in continuous waterbody areas affected by algae and texture variations, models are more susceptible to misclassification. This comparison highlights that most models struggle to maintain segmentation performance in large-scale water scenes without clear boundaries and are easily affected by surface texture interference. In wave reflection scenarios caused by sunlight creating dynamic reflection patterns on water surfaces, segmentation accuracy is significantly impaired. Baseline models tend to overemphasize local texture features and frequently misidentify specular reflection regions as nonwater. Models such as Swin transformer and MSResNet partially suppress such texture noise due to their ability to model long-range dependencies. SPPrompt-Water not only preserves the structural integrity of waterbodies but also substantially reduces false positives in reflective areas. In ice–water mixed scenarios where ice coverage causes heterogeneity on the water surface, model generalization

is seriously challenged. Baseline methods often misclassify regions based on texture features present at the ice–water interface or on the ice surface itself. In contrast, SPPrompt-Water, empowered by a progressive scale-aware prompting mechanism, remains unaffected by the complexity of mixed ice–water areas, demonstrating its technical advantage in handling challenging waterbody segmentation tasks. Furthermore, by incorporating a scale-aware guidance strategy, our method enables the model to focus on the holistic shape characteristics of waterbodies while enhancing semantic understanding through prompts. This approach not only improves the recognition of large, continuous waterbodies but also ensures high segmentation accuracy for fine-grained and small-scale water regions, reflecting the strong adaptability and generalization ability of the proposed approach.

To evaluate the performance of different methods on full-scene remote sensing images, we selected four representative complex scenarios from the CWBSD dataset: lakes with a mix of ice and water, lakes affected by solar reflection, lakes

TABLE III
QUANTITATIVE COMPARISON RESULTS OF GLH

Methods	OA	mIoU	F1score
U-Net	95.47%	72.79%	77.46%
SwinTransformer	96.16%	74.54%	78.60%
QTUNet	95.16%	70.09%	74.47%
MECNet	95.98%	73.96%	77.93%
MSLANet	95.12%	73.26%	77.21%
MSResNET	97.63%	78.14%	81.82%
SkySense++	97.91%	80.12%	82.34%
SPPrompt-Water (ours)	99.09%	86.91%	89.54%

covered by thin clouds, and lakes rich in algae. We then compared our proposed method with the three top-performing baseline methods in the previous quantitative results—Swin transformer, MSResNET, and SkySense++.

As shown in Fig. 10, in the lake with a mix of ice and water scenario, all three baseline methods exhibit a noticeable performance drop, generally being disturbed by the bright textures of snow-covered ice, which leads to confusion in identifying ice-covered regions. In contrast, our method, equipped with a cross-level prompt-guided prior mechanism, effectively suppresses the interference from snow textures, maintaining higher recognition stability in ice-covered areas. In the lake affected by solar reflection scenario, the Swin transformer produces evident mosaic-like artifacts, indicating that it is strongly disturbed by high-frequency reflection patterns, as well as prediction discontinuities commonly observed in sliding-window inference. Intense solar reflection also affects MSResNET and SkySense++ to some extent, causing partial omission in the central regions of the lake, while our method maintains spatially consistent predictions across subregion boundaries.

3) *GLH-Water*: The quantitative comparison results on the GLH dataset are shown in Table III. Methods such as U-Net and QTUNet achieve mIoU values mostly in the 70%–74% range, indicating relatively limited performance. Swin Transformer shows some improvement over other models but remains insufficient for handling complex water body scenarios. Models incorporating multiscale and residual structures, such as MSResNET, further increase the mIoU to 78.14%. The latest vision foundation model in the remote sensing domain, SkySense++, achieves 80.12% mIoU with the benefit of large-scale pre-training. In comparison, SPPrompt-Water achieves the best performance across all metrics, with an mIoU of 86.91%, representing a 6.79% improvement over SkySense++. This significant gain fully demonstrates the effectiveness of the proposed hierarchical prompt-guided framework for complex water body segmentation tasks.

Fig. 11 presents visual comparison results of different methods in representative complex scenarios, providing an intuitive assessment of each method's robustness and generalization ability in real remote sensing environments. In the first column of Fig. 11, small water bodies can still be observed in the gaps between bridges. However, none of the methods except ours were able to identify these narrow water regions. Additionally, large areas of aquatic vegetation are present beneath the

bridges. Both U-Net and QTUNet are significantly affected (see the blue circles), resulting in misclassification of vegetated areas as water. In the third column of Fig. 11, strong thin cloud interference is present. Most methods experience severe segmentation degradation under this condition, with a substantial increase in erroneous regions. Swin Transformer performs relatively better but still exhibits several misclassified areas. In contrast, our method maintains high segmentation accuracy even under these complex interference conditions. In the fourth column of Fig. 11, shadows caused by tall buildings partially blocking sunlight are visible (see the red circles). Except for SkySense++ and our method, all other methods misidentify shadowed regions as water. While SkySense++ can suppress shadow interference to some extent, our method demonstrates the most stable fine-grained distinction between water and shadows.

In summary, our method consistently shows superior discriminative ability and robustness in challenging scenarios, including narrow bridge-gap water bodies, interference from aquatic vegetation, thin cloud coverage, and shadow confusion, further highlighting its practical applicability in complex remote sensing environments.

D. Model Analysis and Limitations

1) *Ablation Studies*: To verify the effectiveness of the prompt guidance module and the enforced prompt-based parallel simulation training, we conducted ablation experiments on the CWBSD dataset using SPPrompt-Water, focusing on the impact of two core components: the prompt mechanism and the training strategy on segmentation performance.

In the first group of ablation experiments, the model structure was kept unchanged, but no prompt images were provided, in order to evaluate performance under the no-prompt condition. In the second group, only prompt images were input, but the enforced-prompt parallel simulation training was not applied. The third group employed the full method proposed in our work.

The results of the ablation experiments shown in Table IV and Fig. 12 indicate that the introduction of the prompt mechanism significantly enhances the model's fine-grained segmentation capability. Adding the prompt mechanism improved the mIoU from 81.43% to 90.04% and the *F1*-score from 84.98% to 92.45%, confirming that the prompt mechanism, by guiding spatial attention, effectively strengthens the model's ability to capture details and accurately identify boundaries of multiscale water bodies. Furthermore, applying the enforced prompt training strategy increased OA by 0.38%, mIoU by 2.23%, and *F1*-score by 1.52%. This performance improvement primarily stems from the enforced prompt parallel simulation training strategy, which effectively guides the model to learn more robust feature representations, thereby enhancing its ability to recognize water regions in complex backgrounds. Through the enforced prompt mechanism in the pretraining phase, the model develops a deep understanding of multiscale water body targets during training and can more precisely capture the spatial distribution characteristics of the target regions.



Fig. 11. Visual results of different methods on GLH. (a) Image. (b) Label. (c) U-Net. (d) Swin transformer. (e) QTUNet. (f) MECNet. (g) MSLANet. (h) MSResNET. (i) SkySense++. (j) SPPrompt-Water (ours).

TABLE IV
QUANTITATIVE COMPARISON RESULTS OF ABLATION STUDIES

SPPrompt-Water		OA	mIoU	F1score
Prompt	Enforced Prompt			
-	-	92.73%	81.43%	84.98%
✓	-	99.13%	90.04%	92.45%
✓	✓	99.51%	92.27%	93.97%

TABLE V
MODEL SIZE AND INFERENCE EFFICIENCY COMPARISON

Methods	Parameters	FLOPS	Time
Unet	31.0M	875.8G	10.98
SwinTransformer-Tiny	41.6M	878.1G	11.55
MECNet	30.1M	742.9G	9.34
QTUNet	31.9M	176.5G	6.3
LLAMNet	34.4M	599.2G	7.51
MSResNet	69.6M	550.1G	7.2
SPPrompt-Water (ours)	132.1M	1250.0G	38.51
SkySense++	347.8 M	9621.1G	113.21

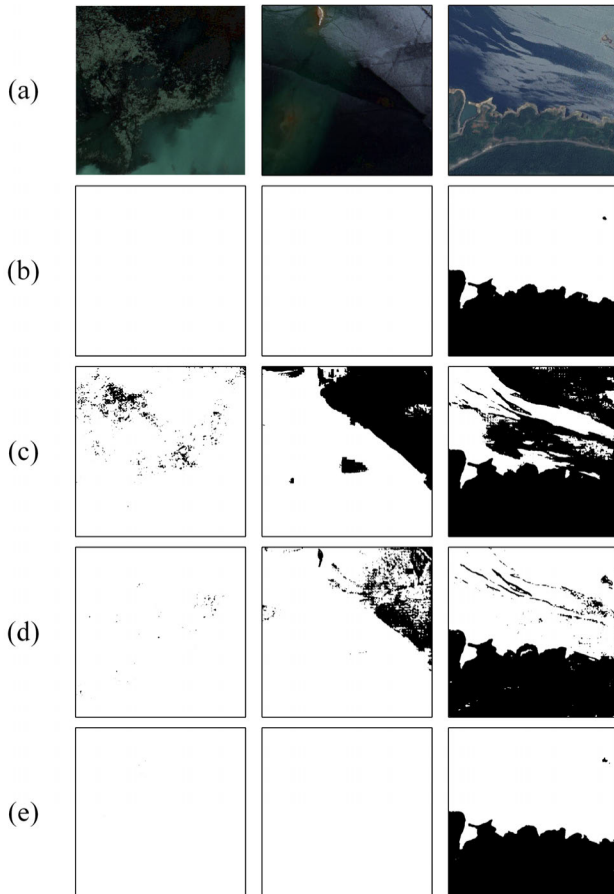


Fig. 12. Visual results of ablation studies. (a) Image. (b) Label. (c) Without prompt. (d) Without enforced prompt. (e) SPPrompt-Water (ours).

2) *Efficiency Analysis*: To comprehensively evaluate the differences in model size and computational efficiency across methods, we further compare seven representative networks in terms of the number of parameters, floating-point operations (FLOPs), and full-scene inference time (in seconds), as shown in Table V. The full-scene inference time is measured on an NVIDIA V100 GPU using 30 images from the GID dataset

(resolution: 7300×6908). The average value is reported to eliminate variations caused by I/O operations.

The results show that SPPrompt-Water contains 132.1 M parameters, requires 1250.0G FLOPs, and achieves a full-scene inference time of 38.51 s. Although these values are slightly higher than those of traditional convolutional architectures such as U-Net and MECNet, SPPrompt-Water performs significantly better than the remote-sensing foundation model SkySense++ (347.8 M parameters, 9621.1G FLOPs, 113.21 s). The increase in computational cost mainly arises from the multiscale prompt guidance and cross-layer feature interaction modules, which require multilevel contextual fusion during inference. However, these components lead to substantial improvements in segmentation accuracy and boundary completeness. Compared with models that rely solely on local convolutions or single-scale Transformers, SPPrompt-Water strikes a favorable balance between accuracy and efficiency, demonstrating strong practical potential and engineering scalability for full-scene segmentation in large-format, high-resolution remote sensing imagery.

3) *Discussion*: Although SPPrompt-Water Net performs well in complex water body scenarios, there are still several considerations for practical applications.

On the one hand, the model's inference relies on multiscale prompt generation and a hierarchical guidance mechanism, which can incur substantial time and memory costs when processing ultra-large images. While our experiments have demonstrated the model's capability to handle full-scene high-resolution imagery, it may still be relatively heavy compared to lighter-weight models in production environments with strict efficiency requirements.

On the other hand, our method is specifically designed to address the challenge of preserving the overall structural continuity of large-scale water bodies in high-resolution remote sensing images. It therefore shows clear advantages when processing water bodies that span large spatial extents, have complex shapes, or exhibit strong upstream-downstream connectivity, such as large rivers and lakes. In contrast, for scenarios dominated by small-scale water bodies, such as ponds or narrow channels, where spatial extent is limited and long-range contextual dependencies are weak, the benefits of multiscale prompt guidance are less pronounced. In these cases, conventional networks can often achieve stable predictions, and the use of more complex prompt mechanisms does not provide significant improvements. Similarly, preliminary experiments on other geospatial classes (e.g., roads) indicate that the proposed prompt-guided strategy retains a certain degree of transferability. However, due to the inherent differences in spatial topology and structural characteristics among different object categories, achieving optimal performance in nonwater scenarios may require adapting the prompt design to class-specific structural priors.

V. CONCLUSION

To address the challenges of multiscale feature variation and texture interference in water body extraction from high-resolution remote sensing images, we propose a scale-progressive prompt-based framework for water extraction. This

method simulates the human “global-to-local” hierarchical cognitive process by designing a hierarchical progressive prompt learning strategy that integrates the global semantic modeling capability of the prompt guidance module with the local detail extraction strengths of the Swin transformer. This enables cross-scale contextual knowledge transfer and noise suppression. On the GID, CWBSD, and GLH-Water datasets, SPPrompt-Water improves mIoU by approximately 8.1%, 6.65%, and 6.79%, respectively, demonstrating clear advantages over the best existing models. Ablation studies further validate the effectiveness of the morphology-driven enforced prompt supervision strategy, which dynamically constrains target region activation and suppresses background noise, significantly enhancing the model’s ability to finely delineate heterogeneous water boundaries and maintain spatial continuity in large-scale water bodies. This work offers new insights for dynamic monitoring of complex water bodies in high-resolution remote sensing imagery.

ACKNOWLEDGMENT

The authors would like to thank the Supercomputing Center of Wuhan University for providing computing power to conduct experiments.

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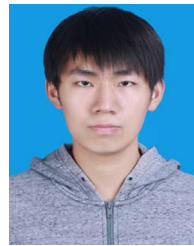
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